

CS223 Computer Architecture & Organization

Instruction Encoding



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Addressing Modes

- ❖ The way in which the operand of an instruction is specified.
 - ❖ Implicit
 - ❖ Immediate
 - ❖ Direct
 - ❖ Indirect
 - ❖ Register
 - ❖ Register Indirect
 - ❖ Indexed
 - ❖ Auto increment/Auto decrement

RISC vs CISC

- ❖ RISC: Reduce the cycles per instruction at the cost of the number of instructions per program.
- ❖ RISC: Simple instruction, simple encoding, fewer addressing modes, more instructions for a task, large code base.
- ❖ CISC: Attempts to minimize the number of instructions per program but at the cost of an increase in the average number of cycles per instruction.
- ❖ CISC: Instructions of varying complexity, variable encoding, more addressing modes, fewer instructions for a task, small code base.

RISC vs CISC

CISC

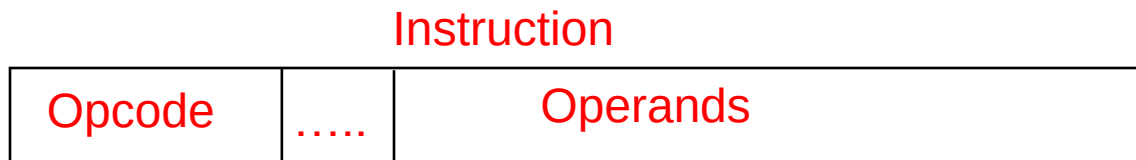
- ⊙ Emphasis on hardware
- ⊙ Multiple instruction sizes and formats
- ⊙ Less registers
- ⊙ More addressing modes
- ⊙ Extensive use of microprogramming
- ⊙ Instructions take a varying amount of cycle time
- ⊙ Pipelining is difficult

RISC

- ⊙ Emphasis on software
- ⊙ Instructions of same set with few formats
- ⊙ Uses more registers
- ⊙ Fewer addressing modes
- ⊙ Complexity in compiler
- ⊙ Instructions take one cycle time
- ⊙ Pipelining is easy

Instruction Formats

- ❖ Layout of bits in an instruction
- ❖ Includes opcode
- ❖ Includes (implicit or explicit) operand(s)
- ❖ Usually more than one instruction format in an instruction set



Allocation of Bits

- ❖ Number of addressing modes
- ❖ Number of operands
- ❖ Register versus memory
- ❖ Number of register sets
- ❖ Address range
- ❖ Address granularity

10010010101010011100010010001010

11110000001010101100010010001010

Instruction Length

- ❖ Affected by and affects:
 - ❖ Memory size
 - ❖ Memory organization
 - ❖ Bus structure
 - ❖ CPU complexity

10010010101010011100010010001010

Example -1

- ❖ A processor has 16 distinct instructions (unique opcodes) and 32 general purpose registers. A 24-bit instruction word has an opcode, two registers operands and an immediate operand. How many bits are possible for the immediate operand field?
- ❖ 16 opcodes: 4 bits for opcode field
- ❖ 32 GPR : 5 bits to address them

Instruction (24 bits)

Opcode	Operand1	Operand2	Imm. Operand
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4	5	5	?
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Example -2

- ❖ A RISC processor has 16 general purpose registers. It operates on 16-bit instruction and has broadly three type of instructions: register type, memory type and branch type. The register type instruction has 3 register operands and memory type instruction has an 8-bit memory offset and a single register operand. The branch instruction has 2 register operands. If there are 8 unique opcodes for register type instructions, and 6 unique opcodes for memory type instruction, what is the maximum number of unique opcodes possible for branch type instructions? Illustrate an instruction encoding pattern.

Example -2

- ❖ 16 GPR = 4 bits.
- ❖ 16-bit instruction: R-Type, M-Type & B-Type
- ❖ R-Type: 4-R+4-R+4-R (8 opcodes), M-Type: 4-R+8-M (6 opcodes)
- ❖ B-Type: 4-R +4-R (opcodes=?)

Instruction (16 bits)

R-Type	0 X X X	R (4)	R (4)	R (4)
M-Type	1 X X X	R (4)	M (8)	
B-Type	1 1 1 0 1 1 1 1	XXXX	R (4)	R (4)

0000 to 0111 (8 opcodes)

1000 to 1101 (6 opcodes)

1110XXXX (16 opcodes) +
1111XXXX (16 opcodes) = 32

Example -3

❖ A processor has 64 registers and it uses a 2-byte instruction format for all its instructions. It has two types of instruction. I-type and R-type. The I-type instruction contains an opcode, a register name and a 4-bit immediate value. The R-type instruction contains an opcode, and two register names. If there are 16 distinct opcodes for I-type instruction, then how many distinct opcodes are possible for R-type instruction? Illustrate the instruction encoding pattern.

❖ 64 GPR = 6 bits.

❖ 16 bit instruction: I-Type & R-Type

❖ I Type: opcode + 6-R + 4-Immediate (16 opcodes)

❖ R Type: opcode+ 6-R + 6-R (max opcodes = ?)

Example -3

- ❖ 64 GPR = 6 bits.
- ❖ 16 bit instruction: I-Type & R-Type
- ❖ I-Type: opcode + 6-R + 4-Immediate (16 opcodes)
- ❖ R-Type: opcode+ 6-R + 6-R (max opcodes = ?)

Instruction (16 bits)

I-Type	0 0 X X X X	Imm (4)	R (6)
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XXXX= 0000 to 1111 (16 ops)

R-Type	0 1 1 0 X X 1 1	R (6)	R (6)
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XX= 00 to 11 (4 ops)

TOTAL = 3X4 =12 opcodes

Example -4

- ❖ A processor with 32 general purpose registers (R0, R1, R2..., R31) and 64 multimedia registers (M0, M1, M2, ..., M63) uses fixed length instructions that are 2 bytes long. The processor operates in two modes; normal mode and multimedia mode. Instructions are encoded such that if msb of the instruction is 0 it indicates a normal mode instruction else it is a multimedia instruction. Normal mode has 2 types of instructions; A and B. There are eight type-A instructions, each with two general purpose register operands. Each type-B instruction has an 8-bit immediate value and a general purpose register as its operands. Each multimedia instruction has two multimedia register operands.
 - (a) What is the maximum number of type-B instructions and multimedia instructions this processor can support?
 - (b) Illustrate an instruction encoding pattern that satisfies the above specifications.

Example -4

- ❖ Instructions Length = 2 byte = 16 bits
- ❖ Normal Mode => 0
- ❖ A: 1 bit (reserved to find out normal/multimedia) + 5 + 5 => $16 - 11 = 5$ bits
- ❖ B: 1 bit (reserved to find out normal/multimedia) + 8 + 5 => $16 - 14 = 2$ bits

Instruction (16 bits)

0 [Normal]	0 0 X X X [A]	Reg-5	Reg-5
0 [Normal]	01 [B] 10 11	Imm-8	Reg-5
1 [MM]	X X X	MMReg-6	MMReg-6



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